CCNA 200-301 V2.0

09 Edge-Host Connectivity: PoE, Voice, APs and IoT

Part II — Switching and Network Access

EXAM TOPICS

2.2

LECTURER

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THE COURSE

Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	**Switching and Network Access**	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

LEARNING OUTCOMES

What you will be able to do

- **Configure** an access port correctly for each of the five device types the blueprint names.
- **Explain** how a data VLAN and a voice VLAN share one cable, and why that is not a trunk in the ordinary sense.
- **Configure** and verify Power over Ethernet, including budget and per-port limits.
- **Diagnose** the faults peculiar to edge ports: a phone that gets no power, an AP that sees only one VLAN, and a device in the wrong VLAN because it was plugged into the wrong socket.

ASSUMED KNOWLEDGE

Before we start

Chapter 8 for VLANs and access ports. Chapter 7 for interface ranges.

KEY TERMS

Vocabulary for this chapter

- Power over Ethernet
- PoE, PoE+, UPoE
- power budget
- voice VLAN
- data VLAN
- standalone AP
- controller-based AP
- LWAPP/CAPWAP
- IoT appliance
- errdisable

Each is defined where it first matters — not here.

ON THE EXAM

Read topic 2.2 carefully — it lists the devices

This is what the blueprint actually asks for

Topic **2.2** is “configure Layer 2 switch port attributes for edge-host connectivity (VLAN, PoE, port channel, and LACP)” and then names five categories: desktops, printers and IoT appliances; wireless access points, standalone and controller-based; VoIP phones; virtualised hosts; and network appliances. The exam expects a different port configuration for several of them, so learn them as five cases rather than one.

Power over Ethernet

DEFINITION

PoE

Power over Ethernet delivers electrical power over the same twisted pair that carries data, so a phone, camera or access point needs one cable rather than two. The switch negotiates with the device and supplies only what it asks for.

WATCH OUT

The budget is the constraint, not the port

A switch has a total power budget shared across all ports — often far less than the number of ports times the maximum per port. A 24-port PoE+ switch with a 370 W budget cannot deliver 30 W to all 24. When the budget is exhausted the switch stops powering *new* devices, so the symptom is “the phone I plugged in this morning works, the one I plugged in this afternoon does not”, and nothing in the port configuration looks wrong.

Check `show power inline` before you check anything else.

CONFIGURE

PoE on an access port

IOS · configuration mode

```
interface GigabitEthernet0/10
 description AP in the east corridor
 switchport mode access
 switchport access vlan 10
 ! auto is the default: negotiate and give the device what it asks for.
 power inline auto
 ! Cap a port that must never take more than a phone's worth.
 ! power inline auto max 15400
 no shutdown
!
! Never power this port -- a data-only port that must not energise.
interface GigabitEthernet0/24
 power inline never
```

show power inline

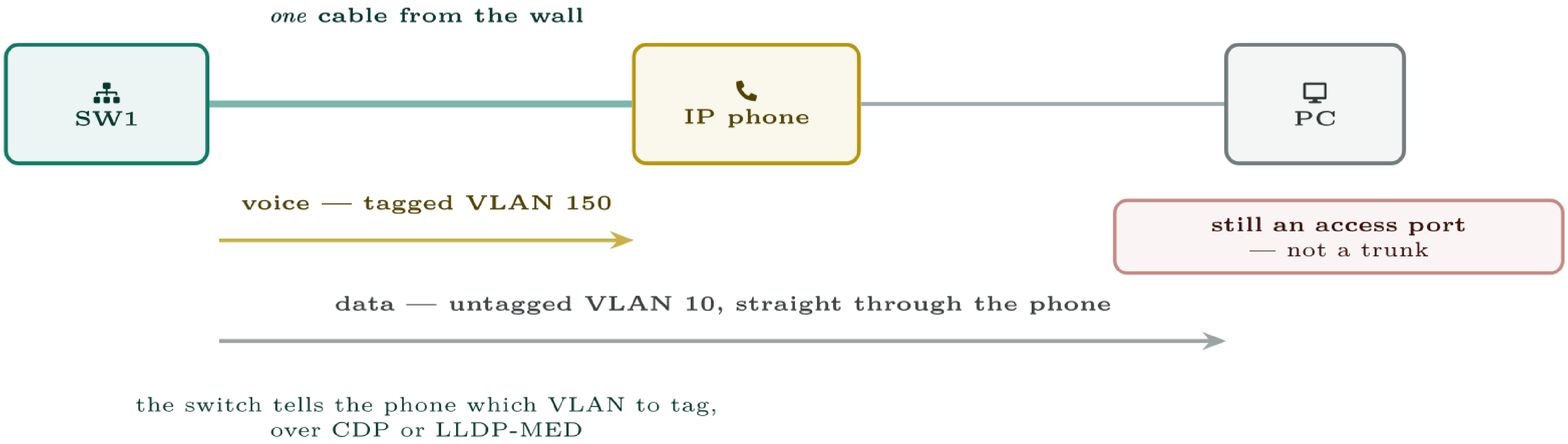
```
SW1#

Available:370.0(w)  Used:71.5(w)  Remaining:298.5(w)

Interface Admin  Oper       Power   Device              Class Max
                            (Watts)
--------- ------ ---------- ------- ------------------- ----- ----
Gi0/1     auto   on         6.3     IP Phone 7960       2     15.4
Gi0/2     auto   on         6.3     IP Phone 7960       2     15.4
Gi0/10    auto   on         28.6    AIR-AP2802I         4     30.0
Gi0/11    auto   off        0.0     n/a                 n/a   30.0
Gi0/24    never  off        0.0     n/a                 n/a   0.0
```

Voice VLANs: two VLANs, one cable

Voice VLANs: two VLANs, one cable



One cable, two VLANs, and no trunk. The phone tags its own traffic and passes the PC's through untouched.

CONCEPT

How one access port carries two VLANs

The port is configured with a **data VLAN** and a **voice VLAN**. The switch tells the phone, over CDP or LLDP-MED, which VLAN ID to tag its own traffic with. The phone tags voice frames and passes the PC’s frames through untagged.

So the switch sees tagged voice traffic and untagged data traffic on one access port. Strictly this is a trunk carrying two VLANs, but it is configured as an access port and Cisco calls it a **multi-VLAN access port**. Do not configure `switchport mode trunk` here — the voice VLAN command does what is needed, and a real trunk would change how the PC’s untagged traffic is handled.

CONFIGURE

A desk port for a phone with a PC behind it

IOS · configuration mode

```
interface GigabitEthernet0/1
 description Desk 1 - phone with PC daisy-chained
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 30
 ! The PC should forward immediately rather than waiting for
 ! spanning tree -- see chapter on Rapid PVST+.
 spanning-tree portfast
 power inline auto
 no shutdown
```

VERIFY

show interfaces GigabitEthernet0/1 switchport

```
SW1#

Name: Gi0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 10 (Staff)
Voice VLAN: 30 (Voice)
```

Access points, standalone and controller-based

WATCH OUT

The most common AP misconfiguration

Configuring a lightweight AP's port as a trunk, or a standalone AP's port as an access port. Both leave the AP reachable — it will get an address and appear healthy — while clients on some or all SSIDs get nothing. If wireless users on one SSID work and another SSID does not, look at the switch port before you touch the wireless configuration.

Both kinds of access point

IOS · configuration mode

```
! Lightweight AP: one access port in the AP management VLAN.
! Client traffic is tunnelled to the controller, so no client
! VLANs need to exist on this switch at all.
interface GigabitEthernet0/10
 description Lightweight AP - CAPWAP to WLC
 switchport mode access
 switchport access vlan 50
 power inline auto
 spanning-tree portfast
!
! Standalone AP: a trunk, because the AP bridges each SSID
! into a different VLAN and tags them itself.
interface GigabitEthernet0/11
 description Standalone AP - SSIDs in VLANs 10 and 20
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 50
 switchport trunk allowed vlan 10,20,50
 power inline auto
```

Printers, IoT appliances and virtualised hosts

Common pitfalls

- **PortFast on a port that leads to a switch.** PortFast skips the spanning tree listening and learning states. On an edge port that is correct; on a port leading to another switch it can create a loop. Use BPDU guard alongside it (Chapter 12).
- **Assuming the phone’s PC port is in the voice VLAN.** It is not — untagged traffic from the PC lands in the data VLAN, which is the whole point.
- **Ignoring the power class.** A class 4 device on a switch with only 802.3af ports will not power up, however correct the configuration looks.
- **Trunking to a lightweight AP.** See the warning above.

Half the wireless users work; the other half get no address.

Users on the `Staff` SSID are fine. Users on `Guest` associate to the wireless network but never receive an IP address. The AP is reachable and its configuration has not changed.

Half the wireless users work; the other half get no address.

What would you check first — and what do you expect to see?

show interfaces GigabitEthernet0/11 trunk

```
SW1#

Port        Mode         Encapsulation  Status        Native vlan
Gi0/11      on           802.1q         trunking      50

Port        Vlans allowed on trunk
Gi0/11      10,50
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. This is a standalone AP, correctly trunked, but the allowed VLAN list carries 10 and 50 only. VLAN 20 — the Guest SSID's VLAN — is not permitted on the trunk, so Guest clients associate at the radio (which the AP handles alone) and then have no path to the DHCP server. Association succeeding while addressing fails is the signature of a VLAN that does not reach the wire.

And the lesson

Fix. Add the missing VLAN to the trunk: `switchport trunk allowed vlan add 20`. Note `add` — without it you replace the whole list and drop VLANs 10 and 50, taking down the users who were working. Confirm with `show interfaces trunk`, then have a Guest client renew.

Wire an edge properly

Topology. One PoE switch; an IP phone with a PC behind it; a lightweight AP; a printer.

1. Create VLANs 10 (Staff), 20 (Guest), 30 (Voice), 50 (AP-Mgmt) and 60 (Printers).
2. Configure the phone port with data VLAN 10, voice VLAN 30, PortFast and PoE. Verify both VLANs appear on the one port.
3. Configure the AP port as an access port in VLAN 50 with PoE.
4. Configure the printer port in VLAN 60 with PortFast.
5. Run `show power inline` and record the budget, the used figure and the class of each powered device.
6. Set `power inline never` on the printer port and observe what happens to a PoE device moved there.

Wire an edge properly (continued)

7. **Extension.** Change the AP to standalone: reconfigure its port as a trunk carrying VLANs 10, 20 and 50, and explain in one sentence why the two AP types need opposite port configurations.

CHECKPOINT 1 of 3

An IP phone powers up and works; a PC plugged into the back of it gets an address in VLAN 10. Which VLAN carries the phone's own traffic, and how did the phone learn to use it?

Discuss · then advance

ANSWER

Checkpoint 1 of 3

An IP phone powers up and works; a PC plugged into the back of it gets an address in VLAN 10. Which VLAN carries the phone's own traffic, and how did the phone learn to use it?

> The voice VLAN, tagged. The phone learned its VLAN ID from CDP or LLDP-MED advertised by the switch. The PC's traffic stays untagged in the data VLAN, which is why one cable carries both.

A new AP will not power on. `show power inline` shows the port `auto/off` and `Remaining:4.2(w)`. What is wrong?

Discuss · then advance

ANSWER

Checkpoint 2 of 3

A new AP will not power on. `show power inline` shows the port `auto/off` and `Remaining:4.2(w)`. What is wrong?

> The switch has only 4.2 W of power budget left, which is not enough for the AP. This is a *budget* problem, not a port problem — the port is willing (`auto`) and the supply cannot meet the request. Add a power supply, move the AP, or free capacity elsewhere.

CHECKPOINT 3 of 3

Why does a lightweight AP need only one VLAN on its switch port when it serves four SSIDs?

Discuss · then advance

ANSWER

Checkpoint 3 of 3

Why does a lightweight AP need only one VLAN on its switch port when it serves four SSIDs?

> Because it tunnels all client traffic to the wireless controller inside CAPWAP. The SSID-to-VLAN mapping is done by the controller, so the switch port only needs to reach the controller. A *standalone* AP bridges locally and does need a trunk.

What to take away

- PoE classes: 15.4 W af, 30 W at, up to 90 W bt. The shared *budget* is what actually runs out, so read `show power inline` first.
- A voice VLAN plus a data VLAN on one access port lets a phone and a PC share a cable: the phone tags, the PC does not.
- Lightweight APs take an access port in the management VLAN; standalone APs take a trunk carrying every SSID's VLAN. Getting this backwards breaks some SSIDs and not others.
- Printers and IoT devices belong in their own VLAN with port security. Virtualised hosts usually need a trunk.
- Use `switchport trunk allowed vlan add`, never bare `allowed vlan`, on a live trunk.

Where we got to

- PoE classes: 15.4 W af, 30 W at, up to 90 W bt. The shared *budget* is what actually runs out, so read `show power inline` first.
- A voice VLAN plus a data VLAN on one access port lets a phone and a PC share a cable: the phone tags, the PC does not.
- Lightweight APs take an access port in the management VLAN; standalone APs take a trunk carrying every SSID’s VLAN. Getting this backwards breaks some SSIDs and not others.
- Printers and IoT devices belong in their own VLAN with port security. Virtualised hosts usually need a trunk.

NEXT

Chapter 10 — Trunking, 802.1Q and SVIs